OS Lecture 1 - Solved Questions

Q1: Select correct answer for the following:

1- According to your knowledge, what is the primary role of an operating system?

A. Translating high-level programming code directly into machine code binaries.

B. Manufacturing physical CPU cycles and hardware memory chips.

C. Serving strictly as an application program for end-user task management.

D. Acting as an intermediary interface between the computer user and the physical hardware.

2- Which of the following characteristics distinguishes system software from application software?

A. It is designed strictly for simple end-user activities such as web browsing and editing documents.

B. It is written exclusively in high-level languages like Python, Java, and C++.

C. It completely removes the need for operating systems and hardware compilers.

D. It is typically written in low-level languages like assembly language and sits closer to computer hardware.

3- In operating system architecture, which component serves as the central core responsible for major operations and direct hardware interaction?

A. Command Interpreter.

B. Application Program.

C. Kernel.

D. Shell.

4- What is the primary function of a shell in an operating system architecture?

A. Compiling source code written in C++ into machine-level assembly instructions.

B. Allocating physical RAM and scheduling CPU instruction threads directly on the silicon processor.

C. Managing physical disk storage platters and BIOS firmware initialization.

D. Providing a command-line or graphical interface for users and applications to interact with the operating system.

5- MS-DOS is presented in the text as a classic example of which operating system architecture?

A. Micro-Kernel Architecture.

B. Simple Architecture.

C. Exo-Kernel Architecture.

D. Layered Architecture.

6- What is a major structural advantage of Micro-Kernel Architecture over Monolith Architecture?

A. Complete removal of device drivers from the system software hierarchy.

B. Simpler initial design and straightforward implementation compared to single-kernel models.

C. Maximum raw performance due to eliminating communication overhead between system components.

D. Increased reliability and stability, as the failure of one microkernel does not bring down the entire system.

7- What was the primary design objective of the Exo-Kernel Architecture developed at MIT?

A. Keeping the kernel size minimal while enabling application programs to manage hardware resources directly.

B. Combining all OS services into a single large piece of central code to streamline banking transactions.

C. Hiding all hardware operations inside hierarchical layers where each upper layer relies on the lower layer.

D. Restricting hardware execution so that only assembly language programs can run on the machine.

8- How are components structured within a Layered Architecture operating system?

A. Hardware forms the bottom layer, the user interface sits at the topmost layer, and intermediate layers hide internal operations from upper layers.

B. Multiple independent kernels execute simultaneously without any hierarchical ordering.

C. Application programs sit at the bottom layer to give hardware direct control over user inputs.

D. All operating system services are compiled into a non-modular, single-layer monolithic binary.

9- From the system's point of view, what are the two main roles fulfilled by an operating system?

A. High-level language compiler and assembly parser.

B. Hardware manufacturer and physical disk drive cleaner.

C. User interface designer and web browser manager.

D. Resource allocator and control program.

10- Which of the following is identified as a primary disadvantage of Monolith Architecture?

A. It suffers from severe performance degradation due to microkernel communication overhead.

B. It is crash prone and difficult to enhance.

C. It prevents users from running high-level programming languages like C++ and Java.

D. It cannot support time-sharing or multi-programming operational models.

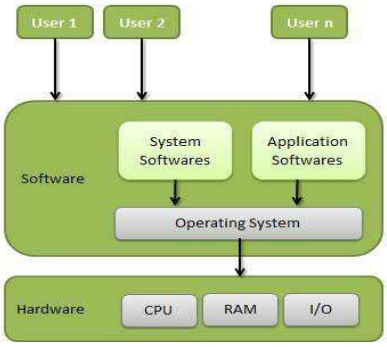
Q2: List only five of OS functions.

- Process Management.
- I/O Device Management.
- File Management.
- Security Management.
- Main Memory Management.

Q3: Define OS and explain with diagram computer system components.

OS Lecture 1 - Solved Questions

An operating system is a program that acts as an interface between the user and the computer hardware and controls the execution of all kinds of programs. OS acts as an intermediary between a user of a computer and the computer hardware.



Computer system can be divided into four components:

1. Hardware, includes Monitor, Keyboard, CPU, Disks, Memory, etc.
2. System Software are the software which are more complex in nature and they are more near to computer hardware. The software is usually written in low-level languages like assembly language and includes Operating Systems (Microsoft Windows, Mac OS, and Linux), Compiler, and Assembler etc.
3. Application programs are the software which users use directly to perform different activities. These softwares are simple and easy to use like Browsers, Word, Excel, different Editors, Games etc. These are usually written in high-level languages, such as Python, Java and C++.
4. Users, whom use the overall computer system.